CD-DSL: Consensus-based Decentralized Distributed Swarm Learning for Edge Intelligence with Heterogeneous Big Data

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***Abstract*—Modern artificial intelligence (AI) increasingly relies on large-scale, distributed, and heterogeneous data collected by wireless edge devices. However, practical edge intelligence remains challenging due to non-convex objectives, distributed data heterogeneity, and privacy concerns. To address these issues, this paper proposes a consensus-based decentralized distributed swarm learning framework, termed CD-DSL for heterogeneous wireless edge networks. Our CD-DSL integrates consensus optimization with particle swarm optimization (PSO), by leveraging neighborhood-aware model alignment strategy together with swarm-oriented exploration and exploitation. The consensus mechanism supports decentralized coordination without raw-data exchange, while the swarm-intelligence inspired updates utilize historical and neighbor-shared experience to enhance exploration to escape local traps, improve robustness to data heterogeneity, and accelerate convergence. We further develop an adaptive neighbor-mixing strategy that learns performance-aware consensus weights using local validation feedback, thereby improving decentralized collaboration among heterogeneous edge devices. Theoretical analysis indicates that CD-DSL maintains participant consistency and achieves non-ergodic convergence to a neighborhood of a stationary point under non-convex objectives. Experimental results show that CD-DSL achieves efficient and robust decentralized learning for edge intelligence with heterogeneous big data.**

***Index Terms*—Big data, heterogeneity, non-convex optimization, distributed swarm learning, decentralized edge network.**

I. Introduction

The rapid growth of edge intelligence in wireless networks is shifting learning and inference from centralized platforms toward large populations of edge devices. This paradigm enables real-time decision-making, scalable connectivity, and low-latency intelligent services across emerging wireless applications [1]. Big data from drones, sensors, vehicles, and mobile terminals are inherently large-scale and highly heterogeneous, leading centralized processing impractical due to the prohibitive overhead for data transmission and computation [2]. In contrast, distributed learning allows collaborations without raw data collection [3]–[5], but suffering performance degeneration from non-identically and independent distributed (Non-IID) input data and non-convex objective at the edge [6].

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Federated learning (FL) has been widely investigated to address data heterogeneity in distributed settings [7]–[9]. Unfortunately, most pioneer FL frameworks hinge on a central server for model aggregation and coordination, which is vulnerable to node/link failures and faces communication bottleneck as well as concerns regarding trust in the central server [10]. These limitations motivate a more robust and efficient decentralization for edge intelligence. However, replacing the central server with peer-to-peer collaboration does not automatically produces effective decentralized learning. Moreover, without a global coordinator, decentralized learning may suffer from slower convergence and weaker agreement guarantees, especially given massive device population with heterogeneous big data under non-convex objectives.

Consensus optimization provides a principled decentralized paradigm for coordination and agreement among distributed devices [11], where consensus mechanisms enable neighboring devices to iteratively exchange model information and reach agreement without relying on a central server. However, classical consensus formulations are mainly developed for convex optimization settings and often assume doubly stochastic weights, which usually does not fit the real-world edge networks. In practical edge networks, heterogeneous device capabilities, asymmetric communication links, and dynamic wireless conditions frequently lead to directed and imbalanced information flows [12]. As a result, conventional consensus-based learning methods may suffer from biased aggregation, degraded convergence, and limited robustness under realistic wireless network conditions. Although existing literature has studied decentralized learning under non-convex objectives [12]–[14], theoretical guarantees and analyses for heterogeneous big data remain largely unattended.

Even worse, data and system heterogeneity are intrinsic characteristics of wireless edge networks, where heterogeneity becomes more challenging in decentralized settings because each device only has a limited local perspective and cannot rely on a central server to provide a common evaluation criterion. Existing decentralized stochastic gradient descent (DSGD) methods primarily focus on improving local update rules or communication efficiency [15], but they often overlook the role of network intelligence in promoting efficient collaboration among heterogeneous participants [16]. As a re-

sult, when local data distributions are highly non-IID, different devices may produce conflicting model updates, resulting in slow convergence, poor generalization, and vulnerability to local traps in non-convex cases.

Our previous work on distributed swarm learning (DSL) suggest a complementary direction to improve efficiency, robustness, and convergence. By integrating AI with biologically inspired swarm intelligence, DSL leverages local historical experience and collaborative users' information to improve learning efficiency under heterogeneous data distributions [17]. However, it is not straightforward to directly apply the DSL framework to decentralized wireless networks. This is because the existing DSL designs often depend on a shared evaluation perspective to score particles or identify the best-performing model, which is not readily available in serverless decentralized systems. They also lack a principled aggregation mechanism for selecting and weighting useful neighbors over directed and heterogeneous communication graphs.

To address these challenges, this paper ...

To reconcile convergence issues in heterogeneous, non-convex tasks, this work extends our previous distributed swarm learning (DSL) [18], [19] to a fully decentralized system, termed consensus based decentralized distributed swarm learning (CD-DSL), for robust and general control of wireless edge devices. CD-DSL performs a win-win combination of consensus optimization and vanilla DSL framework, where the consensus based LDSGD ensures model generalization and the particle swarm optimization (PSO) enhance collaboration efficiency and capacity to escape from local trap. The main contributions of this work are listed as follows:

1) For robustness and privacy concern, we design a BI empowered decentralized learning paradigm CD-DSL to deal with heterogeneous data and non-convex optimization scenarios. The seamless integration of consensus updates and PSO benefits to jointly optimize a general model for Non-IID input without global evaluations and accelerate learning process by historical and neighbors' experience. Experimental results outperforms faster convergence and generalization than state-of-the-art solutions in non-covnex model training process.
2) CD-DSL firstly collaboration scheme introduce advanced network intelligence for heterogeneous edge devices network. Consider realistic wireless environments, we propose an advanced push sum consensus local model updates paradigm for directed structures. The involvement of swarm intelligence benefits for decentralized cooperation to escape from local trap with balanced exploration and exploitation.
3) We demonstrates a comprehensive theoretical analysis to guarantee the convergence and consistency of participants in CD-DSL with only differentiable assumption. Moreover, the consistency proof verify the nonergodic convergence as all the participants of CD-DSL will converge to a common specific neighborhood of the stationary point.

Notations: Let $\mathbf{1} \in \mathbb{R}^{n\times 1}$ denote the all-one column vector and $\mathbf{I} \in \mathbb{R}^{n\times n}$ represent the identity matrix. We apply an event indicator $\mathbb{I}_{(f_{i,t-1}>f_{i,t})}$ to describe the probability of an occurred event $\Pr(y=c)$. Given a matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$, $\|\mathbf{A}\|_2^2$ is the spectral norm. And $\|\mathbf{x}\|_2^2$ denotes the Euclidean norm of a vector $\mathbf{x} \in \mathbb{R}^{n\times 1}$.

II. CD-DSL: Consensus-based Decentralized Distributed Swarm Learning

This section presents the proposed methodology. We first formulate the decentralized learning model, followed by a consensus mechanism for directed topologies [13]. A distributed swarm learning scheme is then introduced to improve cooperation and enhance exploration, along with the overall algorithm.

A. Problem Formulation

Consider a decentralized wireless network with K heterogeneous devices collaboratively performing a shared task. The time-invariant synchronous wireless topology is represented by a strongly connected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with devices set $\mathcal{V}$ and wireless communication links set $\mathcal{E}$. The adjacency matrix $\mathbf{A} = \{a_{i,j}\}_{K\times K}$ characterizes peer-to-peer communication accessibility with the representation $a_{i,j} = 1$ of a reliable communication link from device i to device j. Devices can only reach out their one-hop neighbors for permitted information (raw data query is prohibited). We set $a_{i,i} = 1 \forall i$ for accumulated gains from local historical experiences. Each participants maintains a similar learning model $M_i(\mathbf{w}_i, D_i), \mathbf{w}_i \in \mathbb{R}^{N\times 1}, i = 1, \ldots, K$ with parameter vector $\mathbf{w}_i$ and local observation dataset D_i. However, due to biased local observations and heterogeneous data distributions, independently trained models may lead to suboptimal global solutions. Therefore, the global objective of the UAV system should be formulated to find a general model $\overline{\mathbf{w}}$ to mitigate heterogeneous influence:

$$\overline{\mathbf{w}} \triangleq \arg\min_{\mathbf{w}} \frac{1}{K} \sum_{i=1}^{K} f_i(\mathbf{w}, D_i), \tag{1}$$

where performance f_i is evaluated based on local dataset D_i. Local model updates follow a weighted local decentralized gradient descent mechanism for the generalization of $\overline{\mathbf{w}}$ [20]:

$$\mathbf{w}_{i,t+1} = \sum_{k\in\mathcal{N}_i\cup\{i\}} p_{k,i}\mathbf{w}_{k,t} - \alpha_i \nabla f(\mathbf{w}_{i,t}, D_i), \forall i, \tag{2}$$

where α_i denotes the learning rate of i-th model, $p_{k,i}$ is a trainable parameter representing the alignment of neighboring devices k of device i. $\mathcal{N}_i \cup \{i\}, \mathcal{N}_i = \mathcal{N}_i^{in} \cup \mathcal{N}_i^{out}$ represents the union of neighbors set (in-degree and out-degree neighbors) and the self-loop so that each client can preserve own historical experience and benefit from the neighbor's current training gains. We define a state space model for the global iteration of Markov training process $\mathbf{W}_{t+1} = \mathbf{W}_t \cdot \mathbf{P}_{K\times K} - \Lambda, i = 1, \ldots, K$, here $\mathbf{P} = [\mathbf{P}_i] = \{p_{i,j}\}$ is the mixing matrix to aggregate training gains from neighbors. $\mathbf{P}$ is initialized by the column normalizing $\mathbf{A}$ with $p_{i,j}^t \equiv 0, \forall t$ for all

unconnected node pairs. $\Lambda \triangleq (\alpha_i \nabla f(\mathbf{w}_{i,t}, D_i))_1^K \in \mathbb{R}^{N\times K}$ denotes a stacked matrix with the local gradient. For simple notations, let $f_{i,t}$ represents $f(\mathbf{w}_{i,t}, D_i)$.

B. Push Sum Consensus

Consensus optimization orchestrates a common coordination scheme for the consistency of distributed local training [21], which introduces a consensus constraint to streamline the cooperation mode and guarantee general learning gains across participants [22]. Followed the Eq. 2, the matrix form of the optimization is determined as

$$\begin{aligned} \arg\min_{\mathbf{W}} \frac{1}{K}\mathbf{1}^\top \cdot \mathbf{f}(\mathbf{W},\mathbf{D}) &\triangleq \frac{1}{K}\sum_{i=1}^{K} f_i(\mathbf{w}_i, D_i) \\ s.t. \quad \mathbf{w}_i &= \mathbf{w}_j, \forall (i,j) \in \mathcal{E}, \end{aligned} \tag{3}$$

where the stacked vector $\mathbf{f}(\mathbf{W},\mathbf{D}) \in \mathbb{R}^{1\times K}$ denotes the loss function and gradient of parameter $\mathbf{W} \in \mathbb{R}^{N\times K}$.

The mixing matrix $\mathbf{P}$ performs a cornerstone of the consensus based aggregation scheme. For the convergence perspective, a well-documented assumption represents a doubly stochastic $\mathbf{P}$ to aggregate gains from neighbors within undirected wireless topologies [23]. However, the data and system heterogeneity in edge networks leads to asymmetric interactions, violating the symmetric and doubly stochastic property of $\mathbf{P}$. To this end, we design a robust and general consensus scheme for directed topologies, which is more reliable to formulate distributed heterogeneous input in wireless:

Assumption 1. *(Mixing Matrix): Given a time-invariant strongly connected network topologies $\mathcal{G}$, there exist an optimal mixing matrix $\mathbf{P}^\star = \{p^\star_{i,j}\} \in \mathbb{R}^{K\times K}$ is nonnegative, off-diagonal entries and column stochastic as:*

$$\begin{aligned} \mathbf{1}^\top P &= \mathbf{1} \\ 0 &< p_{i,j}, \forall (i,j) \in \mathcal{E} \\ 0 \leq & p_{i,j} \leq 1. \end{aligned} \tag{4}$$

The eigenvalues of $\mathbf{P}$ follows $1 = \lambda_1(\mathbf{P}) \geq |\lambda_2(\mathbf{P})| \geq \ldots \geq |\lambda_K(\mathbf{P})|$.

The mixing matrix approximates the contribution of individual results to one-hop neighbors. P forms a column stochastic, nonnegative and off-diagonal paradigm to avoid unstable combinations from neighbors [14]. The consensus optimization aims to find a common model to mimic the global optimal model such that all trained model will converge to the same neighborhood of given stationary point $\mathbf{w}_1, \ldots, \mathbf{w}_K, \overline{\mathbf{w}} \in [\mathbf{w}^\star - \epsilon, \mathbf{w}^\star + \epsilon]$.

Although the column stochastic aggregation graph reduces modeling complexity and executes better in directed topological structure, the substituted row stochastic incurs an imperative gap as $\frac{1}{K}\sum_i \mathbf{w}_{i,\infty} \neq \frac{1}{K}\sum_j \mathbf{w}_{j,\infty}$ as $\sum_i p_{i,j} \neq 1$. This gap incurs a biased objective $\overline{\mathbf{w}} = \arg\min_{\mathbf{w}} \frac{1}{K}\sum_{i=1}^{K} \mathrm{Pr}_i\, f_i(\mathbf{w}_i, D_i)$ from the optimal aggregation, where the coefficient Pr_i is attributed to the deviant stationary distribution of $\mathbf{P}$. Inspired by the push sum consensus mechanism [13], we introduce a nonnegative scalar matrix $\mathbf{Z} = \{\mathbf{z}_i\}_1^K, \mathbf{z} \in \mathbb{R}^{N\times 1}$ to rebuild symmetric matrix for global updating. With the initialized $\mathbf{z}_{i,0} = \mathbf{1}, \forall i$, individual parameter updates follow a ratio consensus prototype:

$$\begin{aligned} \mathbf{W}_{t+1} &= (\mathbf{W}_t \cdot \mathbf{P}) \circ (\mathbf{Z}_t \cdot \mathbf{P})^{\circ -1} - \Lambda \\ \mathbf{Z}_{t+1} &= \mathbf{Z}_t \cdot \mathbf{P} = \mathbf{Z}_0 \cdot \mathbf{P} \\ \mathbf{X}_{t+1} &= (\mathbf{W}_{t+1} \circ \mathbf{Z}_{t+1})^{\circ -1}, \end{aligned} \tag{5}$$

where $\circ$ represents the Hadamard product and $A \circ B^{\circ -1}$ denotes the element-wise division of two matrices. $\mathbf{Z}$ reveals the accumulated aggregated gains on the scalar matrix. $\mathbf{X}$ is the ratio of parameter $\mathbf{w}_{j,t}$ and scalar $\mathbf{z}_{j,t}$ for unbiased aggregation. Devices transmit $\mathbf{X}$ as an adjusted individual gains to their neighbors. The receiving devices sum up all the information with column stochastic $\sum_j p_{j,i} = 1$ to promote consistency. Based on Perron-Frobenius [21] validation $\lim_{t\to\infty} \mathbf{W}_t \circ \mathbf{Z}_t^{\circ -1} = \frac{1}{K}\mathbf{1}\mathbf{1}^\top \mathbf{W}_0$, proposed consensus algorithm $\overline{\mathbf{w}}_{t+1} = \frac{1}{K}\sum_i \left(\frac{\mathbf{w}_{i,t}}{\mathbf{z}_{i,t}} - \alpha_i \nabla f(\mathbf{w}_{i,t})\right)$ will converge to a specific neighborhood with proper learning rates and sufficient iteration. Push sum mechanism guarantees the consensus recognition at the deployment of decentralized learning frameworks in directed wireless topologies. Without the row stochastic restrictions, proposed consensus scheme benefits to learn a more flexible $\mathbf{P}$ for the precise and reliable approximation about topological interactions.

Although proposed mechanism guarantees the generalization and nonergodic convergence of decentralized learning paradigm for edge computing, data heterogeneity and nonconvex objectives in wireless networks emerged as a critical root flaw for consensus based decentralized learning algorithm. The realistic deployment requires a robust and efficient distributed learning framework across complex environment.

C. Decentralized Distributed Swarm Learning

The prevalence of non-IID observations, non-convex downstream tasks, and one-hop communication constraints calls for a more robust decentralized learning framework in wireless edge networks. Network intelligence provides an effective way to improve the consistency and generalization of distributed learning under heterogeneous inputs. Adaptive model aggregation can alleviate slow convergence by enriching local decisions with valuable experience from neighboring agents and strengthening exploration for improved modeling simultaneously [12]. Distributed swarm learning (DSL) has demonstrated strong robustness to heterogeneity and enhanced exploration and exploitation in non-convex optimization, making it a natural complement to consensus-based decentralized learning [18], [19]. The learning scheme of vanilla DSL

follows the parameter updates of particle swarm optimization algorithms:

$$\begin{aligned}\mathbf{v}_{i,t+1} =& \mathbf{w}_{i,t+1} - \mathbf{w}_{i,t+1} \\ =& c_0\mathbf{v}_{i,t} + c_1\left(\mathbf{w}_{i.t}^l - \mathbf{w}_{i,t}\right) + c_2\left(\mathbf{w}_t^g - \mathbf{w}_{i,t}\right) \\ & - \alpha\nabla F\left(\mathbf{w}_{i,t}, D_g\right),\end{aligned} \tag{6}$$

where the velocity $\mathbf{v}_{i,t}$ is the increment of model parameters in device i at t communication round, $\mathbf{w}_{i.t}^l$ and $\mathbf{w}_t^g$ denotes the local and global best model for the exploitation of outstanding experience. The inertia hyperparameter c_0 controls the exploration capability of model updating. The coefficients c_1, c_2 regulate the exploitation of local and global experience, respectively. The exploration term $c_0\mathbf{v}_{i,t}$ enables the local model to perform exploratory updates that may temporarily deviate from locally optimal directions, strengthening the ability to escape saddle points and search for globally favorable solutions.

The local best $\mathbf{w}_{i.t}^l$ and global best design $\mathbf{w}_t^g$ serve as the indispensable guidance, enabling each agent to exploit both its historical best solution and valuable experience collected from neighboring participants. Such enhanced network intelligence improves the update efficiency, model generalization and collaborative scheme, in which the PSO-based update mechanism mitigate conflicting training gains with heterogeneous input. However, the one hop communication limits the global horizon of local devices, impeding participants gain achievements out of neighbors in such communication round. We introduce a consensus oriented neighbor best model to record the outstanding training gains from one hop neighbors. The flooding information dissemination theory guarantees a common perspective of model evaluation. The accumulated gains of neighbor best tantamount to global best design in a long term. Based on the weighted parameter matrix $\mathbf{X}$ exploitation term is defined as

$$\begin{aligned}\mathbf{X}_t^l =& \left[\mathbb{I}_{(f_{i,t}>f_{i,t-1})}\right]_{i=1}^K \cdot \mathbf{1} \circ \mathbf{X}_{t-1} \\ &+ \left[\mathbb{I}_{(f_{i,t-1}>f_{i,t})}\right]_{i=1}^K \cdot \mathbf{1} \circ \mathbf{X}_t \end{aligned} \tag{7}$$

$$\mathbf{X}_t^n = \mathbf{X}_t \cdot \mathbf{P}, \tag{8}$$

where the event indicator $b_{i,t} \triangleq \mathbb{I}_{(f_{i,t-1}>f_{i,t})}$ denotes the probability that recent model on device i performs better than previous one. Since the training process follows a Markov chain, $\mathbf{X}_t^l$ records the historical best parameters. Thus, the consensus based decentralized distributed swarm learning follows:

$$\begin{aligned}\mathbf{X}_{t+1} =& \mathbf{X}_t \cdot \mathbf{P} - \Lambda(\mathbf{X}_t) + c_0\mathbf{V}_t \\ &+ c_1\mathbf{B}_t \circ \left(\mathbf{X}_{t-1} - \mathbf{X}_t\right) + c_2\mathbf{X}_t \cdot \left(\mathbf{P} - \mathbf{I}\right),\end{aligned} \tag{9}$$

where the probability vector $\mathbf{B} = \left[\mathbb{I}_{(f_{i,t-1}>f_{i,t})}\right] \cdot \mathbf{1} \in \mathbb{R}^{K\times K}$ represents a binary matrix to formulate accumulated training gains, $\mathbf{V}_t = [\mathbf{v}_{i,t}]_{i=1}^K$ records the exploration velocity of participants to escape from local trap. $\mathbf{I}$ represents the identity matrix. For each local updates:

$$\begin{aligned}\mathbf{x}_{i,t+1} =& (1+c_2)\sum_j p_{j,i}\mathbf{x}_{j,t} - \alpha_i\nabla f_{i,t} + c_0\mathbf{v}_{i,t} \\ &+ c_1\mathbb{I}_{(f_{i,t-1}>f_{i,t})}\left(\mathbf{x}_{i,t-1} - \mathbf{x}_{i,t}\right) - c_2\mathbf{x}_{i,t} \\ =& (1+c_2)\sum_j p_{j,i}\frac{\mathbf{w}_{j,t}}{\mathbf{z}_{j,t}} - \alpha_i\nabla f\left(\frac{\mathbf{w}_{j,t}}{\mathbf{z}_{j,t}}, D_i\right) + c_0\mathbf{v}_{i,t} \\ &+ c_0\mathbf{v}_{i,t} + c_1 b_{i,t}\left(\mathbf{w}_{i,t-1} - \mathbf{w}_{i,t}\right) - c_2\mathbf{w}_{i,t}, \forall i.\end{aligned} \tag{10}$$

By incorporating the neighbor best model $\mathbf{X}_t^n$, the mixing matrix serves not only to enforce consensus among local updates, but also to realize a performance aware neighbor aggregation mechanism. Accordingly, we design a joint optimization formulation to reflect the wireless communication topology and adaptive model comparison under heterogeneous inputs. At communication round t, client i learns an adaptive incoming consensus weight vector $\mathbf{p}_i^t = \{p_{j,i}^t\}_{j=1}^K$ by evaluating neighbor models on its local validation data D_i^{val}. The trainable consensus weights are obtained by solving

$$\begin{aligned}\mathbf{p}_i^t = \arg\min_{\mathbf{p}_i} & \sum_{j\in\mathcal{N}_i\cup\{i\}} \left(p_{j,i}f(\mathbf{w}_{j,t}; D_i) + \tau p_{j,i}\log p_{j,i}\right) \\ \text{s.t.} \quad & \sum_{j=1}^K p_{j,i} = 1, \\ & p_{j,i} = 0,\ \forall j \notin \mathcal{N}_i \cup \{i\}. \\ & p_{j,i} \geq 0,\end{aligned} \tag{11}$$

where $f(\mathbf{w}_{j,t}; D_i)$ measures the evaluation loss of neighbor model $\mathbf{w}_j^t$ on local validation dataset. This cross validation design provides a common evaluation of distributed model for heterogeneity measurement. $\tau > 0$ controls the smoothness of the learned consensus weights. The entropy regularization $\tau\sum_{j\in\mathcal{N}_i\cup\{i\}} p_{j,i}\log p_{j,i}$ avoids unstable hard selection of a single neighbor and preserves collaborative consensus. Based on the standard first-order optimality conditions [15], proposed entropy-regularized linear objective is strictly convex and yield a Gibbs-form closed-form solution as

$$p_{j,i} = \frac{\exp\left(-\frac{f_i(\mathbf{w}_j^t; D_i)}{\tau}\right)}{\sum\limits_{k\in\mathcal{N}_i\cup\{i\}} \exp\left(-\frac{f_i(\mathbf{w}_k^t; D_i)}{\tau}\right)}, \quad j \in \mathcal{N}_i \cup \{i\}. \tag{12}$$

In this way, CD-DSL forms a decentralized learning framework that is better suited to heterogeneous input in non-convex optimization.

D. Algorithm Implementation

This section illustrate the implementation of proposed CD-DSL as Algorithm. 1. Since the peer-to-peer wireless network incurs communication overhead, we design an asynchronous timeline to schedule the model updates and joint optimizations with less communication and computation delay. The training process involves T communication round, each round contains

E epoch for local training. In each communication round, devices i receive training gains from $\mathcal{N}_i \cup \{i\}$ and calculate $\mathbf{P}$. Then, local model updates parameters by consensus based decentralized stochastic gradient descent. Next, devices train local model for E epoch for reliable performance with less optimization and communication overhead, where local updates follows renewed local best and neighbor best parameters under decentralized stochastic gradient descent. Finally, participants send trained model and scalar matrix to neighbors and end current communication round. The asynchronous updating design reduce the computational overhead from joint optimization with intermittent peer-to-peer communication.

Algorithm 1 CD-DSL

Initialization:

Initialize $\overline{\mathbf{w}}_t, \mathbf{w}_{i,t}^l, \mathbf{w}_t^n, \mathbf{z}_t$, t, e, i and T, E, given datasets $\{D_i\}$;

for each communication round $t = 1 : T$ **do**
  at edge devices:$i = 1, \ldots, K$
    receive $\{\mathbf{w}_{j,t}, \mathbf{z}_{i,t}\}$ from neighbors in $\mathcal{N}_i \cup \{i\}$;
    evaluate neighbors' models $f(\mathbf{w}_{j,t}; D_i)$ via local dataset;
    update $\mathbf{P}$ via (12) for consensus and performance based aggregation;
  for each epoch $e = 1 : E$ **do**
      update local and neighbor best via (7) and (8);
      update local model via (10);
      transmit $\{\mathbf{w}_{i,t+1}, \mathbf{z}_{i,t+1}\}$ to neighbors;
  end for
end for

III. Performance Analysis

In this section, we analyze convergence behavior of CD-DSL. We firstly detail a ergodic convergence of global model $\overline{\mathbf{w}}$. And then demonstrate a nonergodic convergence neighborhood for all participants $\mathbf{w}_i$. Noting that the proposed consensus mechanism corrects the bias introduced by column-stochastic mixing, the normalized parameter $\mathbf{x}$ represents a linear combination of $\mathbf{w}$. Thus, the convergence of $\mathbf{w}$ can be derived by $\mathbf{x}$ with bounded scalar $\mathbf{z}$. As such, we discuss the convergence of weighted parameters $\mathbf{x}$.

A. Assumption

Following a similar approach in [19], we adopt the common assumptions apart from Ass. 1 as follows.

Assumption 2. *(Bounded dissimilarity): The deviation of l $\mathbf{x}_i$ and the objective model $\overline{\mathbf{w}}$ is bounded by a constant σ [14]*

$$\frac{1}{K}\sum_{i=1}^{K} \|\nabla f_i(\mathbf{x}_i) - \nabla f(\overline{\mathbf{x}})\|^2 \leq \sigma^2. \tag{13}$$

Bounded dissimilarity assumption benefits in the consensus constraint of data and model heterogeneity.

Assumption 3. *(L-differentiable): The loss function of local models is differentiable. $\forall i = 1, \ldots, K, \exists \{L_i\}_1^K$ so that $\forall \mathbf{w}$, the gradient $\nabla f_i(\mathbf{w}_{i,t})$ bounded by constant L_i that is,*

$$\|\nabla f_i(\mathbf{x}_{i,t+1}) - \nabla f_i(\mathbf{x}_{i,t})\| \leq L_i \|\mathbf{x}_{i,t+1} - \mathbf{x}_{i,t}\|, \tag{14}$$

where L is so called Lipschitz constant. We define the $L_{max} = \max\{L_i\}_i^K$ for the nonergodic convergence analysis of $\mathbf{f}(\mathbf{X}, \mathbf{D})$.

Assumption 4. *(Bounded local gradients): For the DSL framework, the gap between the common parameters $\overline{\mathbf{x}}_{t-1}$ and the local parameters $\mathbf{x}_{i,t}, \forall i, t$ is bounded by [18], [24]*

$$|\overline{\mathbf{x}}_{t-1} - \mathbf{x}_{i,t}| \leq \max_{i \in \{1,\ldots,C\}} \{|\alpha \nabla f_{i.t} + (c_0 - c_1 - c_2)\, \mathbf{v}_{i,t}|\}. \tag{15}$$

Since the exploitation property of SGD and PSO, the upper bound of the local model updating is limited by the learning rate and the PSO parameter to limit the exploration steps.

B. Convergence and Consensus Analysis

Let all assumptions hold. The convergence of common model $\overline{\mathbf{x}}$ are bounded by **Theorem 1**

Theorem 1. *The individual parameter updates function is given by* (10)*. For the global optimal model $w^\star$ and communication round $t = 1, \ldots, T$, given the learning rate $\{\alpha_i\}$, the expected convergence rate is bounded by*

Lemma 1. *Under Assumption 1, the common model update satisfies*

$$\bar{\mathbf{w}}_{t+1} = \bar{\mathbf{w}}_t - \alpha_t \bar{\mathbf{g}}_t + \bar{\boldsymbol{\xi}}_t, \tag{16}$$

where

$$\bar{\mathbf{g}}_t \triangleq \frac{1}{K}\sum_{i=1}^{K} \nabla f_i(\widetilde{\mathbf{w}}_{i,t}), \qquad \bar{\boldsymbol{\xi}}_t \triangleq \frac{1}{K}\sum_{i=1}^{K} \boldsymbol{\xi}_{i,t}. \tag{17}$$

Proof. Since $\mathbf{P}_t$ is column stochastic, we have $\mathbf{1}^\top \mathbf{P}_t = \mathbf{1}^\top$. Summing (10) over $i = 1, \ldots, K$ and dividing by K yields (16). □

Lemma 2. *Under Assumptions 2 and 3,*

$$\|\bar{\mathbf{g}}_t - \nabla F(\bar{\mathbf{w}}_t)\|^2 \leq 2L_{\max}^2 e_t^2 + 2\sigma^2. \tag{18}$$

Proof. By the triangle inequality,

$$\begin{aligned}\bar{\mathbf{g}}_t - \nabla F(\bar{\mathbf{w}}_t) =& \frac{1}{K}\sum_{i=1}^{K} \left(\nabla f_i(\widetilde{\mathbf{w}}_{i,t}) - \nabla f_i(\bar{\mathbf{w}}_t)\right) \\ &+ \frac{1}{K}\sum_{i=1}^{K} \left(\nabla f_i(\bar{\mathbf{w}}_t) - \nabla F(\bar{\mathbf{w}}_t)\right),\end{aligned} \tag{19}$$

Applying Assumption 3 and Jensen's inequality gives

$$\left\| \frac{1}{K}\sum_{i=1}^{K} \left(\nabla f_i(\widetilde{\mathbf{w}}_{i,t}) - \nabla f_i(\bar{\mathbf{w}}_t)\right) \right\|^2 \leq L_{\max}^2 e_t^2. \tag{20}$$

Assumption 2 bounds the heterogeneity term by σ^2. Combining the two terms and applying $(a+b)^2 \leq 2a^2 + 2b^2$ yields (18). □

Theorem 2 (Ergodic convergence of the common model)**.** *Suppose all Assumptions hold. Let the learning rate satisfy*

$$0 < \alpha_t \le \bar{\alpha} \triangleq \min\left\{\frac{1}{4L_{\max}}, \frac{1-\lambda}{2\sqrt{2}L_{\max}}\right\}, \tag{21}$$

where $\lambda \triangleq \sup_t |\lambda_2(\mathbf{P}_t)| < 1$. *Then, for any* $T \ge 1$,

$$\frac{\sum_{t=0}^{T-1} \alpha_t \, \mathbb{E}\|\nabla F(\bar{\mathbf{w}}_t)\|^2}{\sum_{t=0}^{T-1} \alpha_t} \le \frac{4\big(F(\bar{\mathbf{w}}_0) - F_{\inf}\big)}{\sum_{t=0}^{T-1} \alpha_t} + 8\sigma^2 + 8L_{\max}^2 \frac{\sum_{t=0}^{T-1} \alpha_t \, \mathbb{E}[e_t^2]}{\sum_{t=0}^{T-1} \alpha_t} + \frac{8\sum_{t=0}^{T-1} \mathbb{E}\|\bar{\boldsymbol{\xi}}_t\|^2/\alpha_t}{\sum_{t=0}^{T-1} \alpha_t}, \tag{22}$$

where $F_{\inf}$ *is a lower bound of* F.

Proof. By $L_{\max}$-smoothness of F,

$$F(\bar{\mathbf{w}}_{t+1}) \le F(\bar{\mathbf{w}}_t) + \frac{L_{\max}}{2}\|\bar{\mathbf{w}}_{t+1} - \bar{\mathbf{w}}_t\|^2 + \langle \nabla F(\bar{\mathbf{w}}_t), \bar{\mathbf{w}}_{t+1} - \bar{\mathbf{w}}_t \rangle \tag{23}$$

Substituting (16),

$$F(\bar{\mathbf{w}}_{t+1}) \le F(\bar{\mathbf{w}}_t) - \alpha_t \langle \nabla F(\bar{\mathbf{w}}_t), \bar{\mathbf{g}}_t \rangle + \big\langle \nabla F(\bar{\mathbf{w}}_t), \bar{\boldsymbol{\xi}}_t \big\rangle + \frac{L_{\max}}{2}\| - \alpha_t \bar{\mathbf{g}}_t + \bar{\boldsymbol{\xi}}_t\|^2. \tag{24}$$

Using

$$\bar{\mathbf{g}}_t = \nabla F(\bar{\mathbf{w}}_t) + \big(\bar{\mathbf{g}}_t - \nabla F(\bar{\mathbf{w}}_t)\big),$$

and Young's inequality, together with $\alpha_t \le 1/(4L_{\max})$, we obtain

$$F(\bar{\mathbf{w}}_{t+1}) \le F(\bar{\mathbf{w}}_t) - \frac{\alpha_t}{4}\|\nabla F(\bar{\mathbf{w}}_t)\|^2 + 2\alpha_t\|\bar{\mathbf{g}}_t - \nabla F(\bar{\mathbf{w}}_t)\|^2 + \frac{2}{\alpha_t}\|\bar{\boldsymbol{\xi}}_t\|^2. \tag{25}$$

Applying Lemma 2,

$$F(\bar{\mathbf{w}}_{t+1}) \le F(\bar{\mathbf{w}}_t) - \frac{\alpha_t}{4}\|\nabla F(\bar{\mathbf{w}}_t)\|^2 + 4\alpha_t L_{\max}^2 e_t^2 + 4\alpha_t\sigma^2 + \frac{2}{\alpha_t}\|\bar{\boldsymbol{\xi}}_t\|^2. \tag{26}$$

Taking expectations, summing from $t = 0$ to $T-1$, telescoping the left-hand side, and dividing by $\sum_{t=0}^{T-1} \alpha_t$ yield (22). □

Theorem 3 (Consensus with bounded bias)**.** *Suppose all Assumptions hold. Then the consensus error defined in* $e_t^2 \triangleq= \frac{1}{K}\sum_{i=1}^{K}\|\widetilde{\mathbf{w}}_{i,t} - \bar{\mathbf{w}}_t\|^2$. *satisfies*

$$e_{t+1} \le \left(\lambda + \sqrt{2}L_{\max}\alpha_t\right) e_t + \sqrt{2}\sigma\alpha_t + \beta_t. \tag{27}$$

Consequently, if

$$0 < \alpha_t \le \frac{1-\lambda}{2\sqrt{2}L_{\max}}, \tag{28}$$

and define

$$\rho_t \triangleq \lambda + \sqrt{2}L_{\max}\alpha_t < 1, \tag{29}$$

then

$$e_t \le \left(\prod_{s=0}^{t-1}\rho_s\right) e_0 + \sum_{s=0}^{t-1}\left(\prod_{\ell=s+1}^{t-1}\rho_\ell\right)\left(\sqrt{2}\sigma\alpha_s + \beta_s\right). \tag{30}$$

Proof. From (10),

$$\widetilde{\mathbf{W}}_{t+1} = \widetilde{\mathbf{W}}_t\mathbf{P}_t - \alpha_t\mathbf{G}_t + \boldsymbol{\Xi}_t, \tag{31}$$

where

$$\mathbf{G}_t \triangleq \left[\nabla f_1(\widetilde{\mathbf{w}}_{1,t}), \ldots, \nabla f_K(\widetilde{\mathbf{w}}_{K,t})\right], \qquad \boldsymbol{\Xi}_t \triangleq \left[\boldsymbol{\xi}_{1,t}, \ldots, \boldsymbol{\xi}_{K,t}\right].$$

Multiplying both sides by

$$\mathbf{I} - \frac{1}{K}\mathbf{1}\mathbf{1}^\top,$$

we obtain

$$\mathbf{E}_{t+1} = \widetilde{\mathbf{W}}_t\left(\mathbf{P}_t - \frac{1}{K}\mathbf{1}\mathbf{1}^\top\right) - \alpha_t\mathbf{G}_t\left(\mathbf{I} - \frac{1}{K}\mathbf{1}\mathbf{1}^\top\right) + \boldsymbol{\Xi}_t\left(\mathbf{I} - \frac{1}{K}\mathbf{1}\mathbf{1}^\top\right). \tag{32}$$

Using the spectral property of $\mathbf{P}_t$,

$$\left\|\mathbf{P}_t - \frac{1}{K}\mathbf{1}\mathbf{1}^\top\right\|_2 \le \lambda,$$

we have

$$e_{t+1} \le \lambda e_t + \alpha_t\left(\frac{1}{K}\left\|\mathbf{G}_t\left(\mathbf{I} - \frac{1}{K}\mathbf{1}\mathbf{1}^\top\right)\right\|_F^2\right)^{1/2} + \left(\frac{1}{K}\left\|\boldsymbol{\Xi}_t\left(\mathbf{I} - \frac{1}{K}\mathbf{1}\mathbf{1}^\top\right)\right\|_F^2\right)^{1/2}. \tag{33}$$

By Assumptions 2 and 3,

$$\frac{1}{K}\left\|\mathbf{G}_t\left(\mathbf{I} - \frac{1}{K}\mathbf{1}\mathbf{1}^\top\right)\right\|_F^2 \le 2L_{\max}^2 e_t^2 + 2\sigma^2, \tag{34}$$

and by Assumption 4,

$$\frac{1}{K}\left\|\boldsymbol{\Xi}_t\left(\mathbf{I} - \frac{1}{K}\mathbf{1}\mathbf{1}^\top\right)\right\|_F^2 \le \beta_t^2. \tag{35}$$

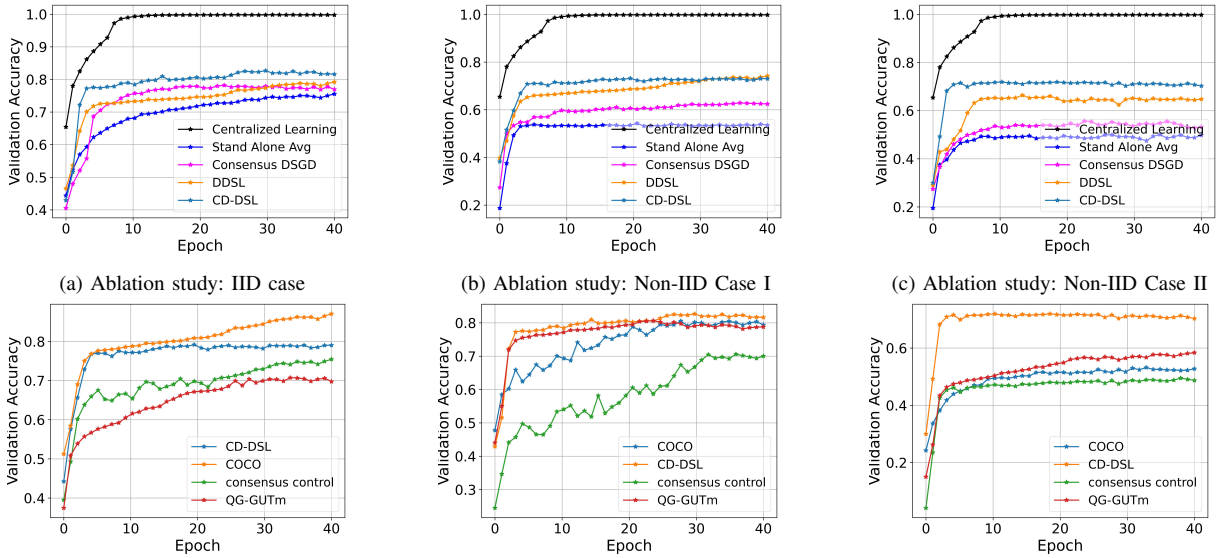
Substituting the two bounds gives (27). Iterating the recursion yields (30). □

Corollary 1 (Constant stepsize neighborhood)**.** *Let* $\alpha_t \equiv \alpha$ *and* $\beta_t \le \bar{\beta}$. *If*

$$0 < \alpha < \min\left\{\frac{1}{4L_{\max}}, \frac{1-\lambda}{2\sqrt{2}L_{\max}}\right\}, \tag{36}$$

then CD-DSL converges to a neighborhood of a stationary point and

$$\limsup_{t\to\infty} e_t \le \frac{\sqrt{2}\sigma\alpha + \bar{\beta}}{1 - \lambda - \sqrt{2}L_{\max}\alpha}. \tag{37}$$



(a) Ablation study: IID case (b) Ablation study: Non-IID Case I (c) Ablation study: Non-IID Case II

(d) Comparison study: IID Case (e) Comparison study: Non-IID Case I (f) Comparison study: Non-IID Case II

Fig. 1: Performance evaluation of proposed CD-DSL. The first row reports the ablation study about CD-DSL and its key components. The second row reports the comparison study against representative decentralized baselines.

Moreover,

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla F(\bar{\mathbf{w}}_t)\|^2 = O\left(\frac{1}{\alpha T}\right) + O(\sigma^2) + O\left(\frac{(\alpha\sigma+\bar{\beta})^2}{(1-\lambda-\sqrt{2}L_{\max}\alpha)^2}\right). \tag{38}$$

Corollary 2 (Diminishing stepsize rate)**.** *Let*

$$\alpha_t = \frac{\alpha_0}{\sqrt{T}}, \qquad \beta_t \le \bar{\beta}\alpha_t, \tag{39}$$

with α_0 *satisfying* (21)*. Then CD-DSL achieves*

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla F(\bar{\mathbf{w}}_t)\|^2 = O(T^{-1/2}), \tag{40}$$

and the consensus bias obeys

$$e_t = O(T^{-1/2}) + O(\rho^t), \qquad \rho < 1. \tag{41}$$

Remark. The above results establish that the common model $\bar{\mathbf{w}}_t$ converges to a neighborhood of a stationary point within non-convex settings. The neighborhood size is jointly determined by the heterogeneity level σ, the network mixing quality λ, the learning rate α, and the swarm hyperparameter β_t.

IV. Simulation Results

To complement the theoretical analysis, we simulates a distributed system and design computational experiments of image classification for CIFAR10 with different settings [25]. We focus on the non-convex parameter optimization tasks as cross entropy loss $\mathcal{L}_{\mathrm{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\log p_{i,y_i}$.

A. Experimental Settings

We design a decentralized systems in Linux 20.04.6 operating system with 4 **Tesla V100-SXM2** GPUs to simulate the collaborative learning across $K = 50$ edge devices[1]. Each contains a individual dataset $|D_i| = 512$ and a validation dataset $|D_i^v| = 128$. Distributed data is deployed on a strongly connected wireless topology with given connection rate 0.7. The heterogeneous data generation relies on the Dirichlet distribution with concentration parameter α. The sampled training and validation datasets share different label distribution across distributed elements. We design a IID case and two Non-IID cases for each tasks with $\alpha = 0.5$ and more diverse case II: 20 sets sampled by $\alpha = 0.1$, 15 sets by $\alpha = 0.5$, 10 sets by $\alpha = 1$ and 5 sets by $\alpha = 10$. ResNet18 [26] is employed for CIFAR10 classification.

For the hyperparameter settings, we apply the SGD optimizer with attenuated learning rate $\alpha_{init} = 0.01, \gamma = 0.5$. The training process involves 40 communication rounds on CIFAR10, each round includes 6 epochs. The batch size is 32 for stable training. The hyperparameters of PSO updates follows $c_0 \sim U\left(0, \frac{0.05}{t}\right)$ for controlled exploration and $c_1, c_2 \sim \mathcal{N}(0, 0.3)$ for balanced exploitation. The consensus weights in joint optimization denotes $\tau = 0.09$.

B. Main Results

Fig. 1 summarizes the validation accuracy of CD-DSL and relevant models under the IID case, Non-IID Case I, and the more challenging Non-IID Case II. The ablation results verify that the proposed CD-DSL consistently outperforms its individual components across all data regimes. In the IID setting, all decentralized variants achieve stable training, the full model reaches the highest validation accuracy and

[1]The code is available at https://github.com/zhuoyu-3/CD-DSL.

converges faster than consensus based DSGD, indicating the involvement of PSO benefits in enhanced experience exploitation. As data heterogeneity increases, the benefit of the proposed integration becomes more evident. In Non-IID Case I, CD-DSL preserves both rapid convergence and strong final accuracy. Although DDSL remains competitive in the early stage, its convergence saturates earlier than CD-DSL, suggesting that swarm intelligence alone is insufficient to maintain consistent collaboration without the proposed consensus design. In the more heterogeneous Non-IID Case II, the gap further widens. CD-DSL remains the best-performing decentralized approach, while the other ablated variants suffer from slower convergence and lower final accuracy. These results validates that consensus optimization improves model alignment and stability, while DSL strengthens exploration and exploitation in heterogeneous settings. The integrated CD-DSL yields gains beyond either component alone.

The comparison results further demonstrate the effectiveness of CD-DSL against SOTA decentralized baselines. Although CD-DSL do not perform better than COCO [8], a decentralized FL framework in the IID setting, the advantage of CD-DSL becomes more pronounced with heterogeneous input. In Non-IID Case I, CD-DSL maintains the similar validation accuracy as the IID case, outperforms most of the training process and converges more stably than the competing methods. The performance gap from consensus control [14] reveals consensus algorithm is insufficient for complex data distributions. CoCo and QG-GUTm remain competitive, but they convergence more slower and the final performance is still lower than that of CD-DSL. With highly imbalanced and statistically diverse Non-IID Case II, CD-DSL preserves a clear margin over all baselines in both convergence speed and final accuracy while the competing methods show either early saturation or unstable improvement. Overall, both the ablation and comparison studies confirm that the integration of push-sum consensus, joint optimization, and DSL-based swarm updates achieves outperforming convergence speed, model consistency, and robustness to heterogeneous big data.

V. Conclusion

This paper proposed CD-DSL, a consensus-based decentralized distributed swarm learning framework for heterogeneous edge intelligence. CD-DSL combines push-sum consensus, adaptive neighbor mixing, and PSO-inspired updates to enable serverless decentralized learning over directed topologies without raw-data sharing. Both theoretical and experimental results showed that CD-DSL achieves robust collaboration under non-IID data and non-convex optimization, with bounded-bias consensus and convergence to a neighborhood of a stationary point. Future work will focus on extending CD-DSL to dynamic wireless environments and streaming data settings for practical applications.

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